OPEN-SOURCE PDF FLIPBOOK

SdFlip

A field guide to turning pages on the web, the way paper does.



Edition 1.0

MIT License

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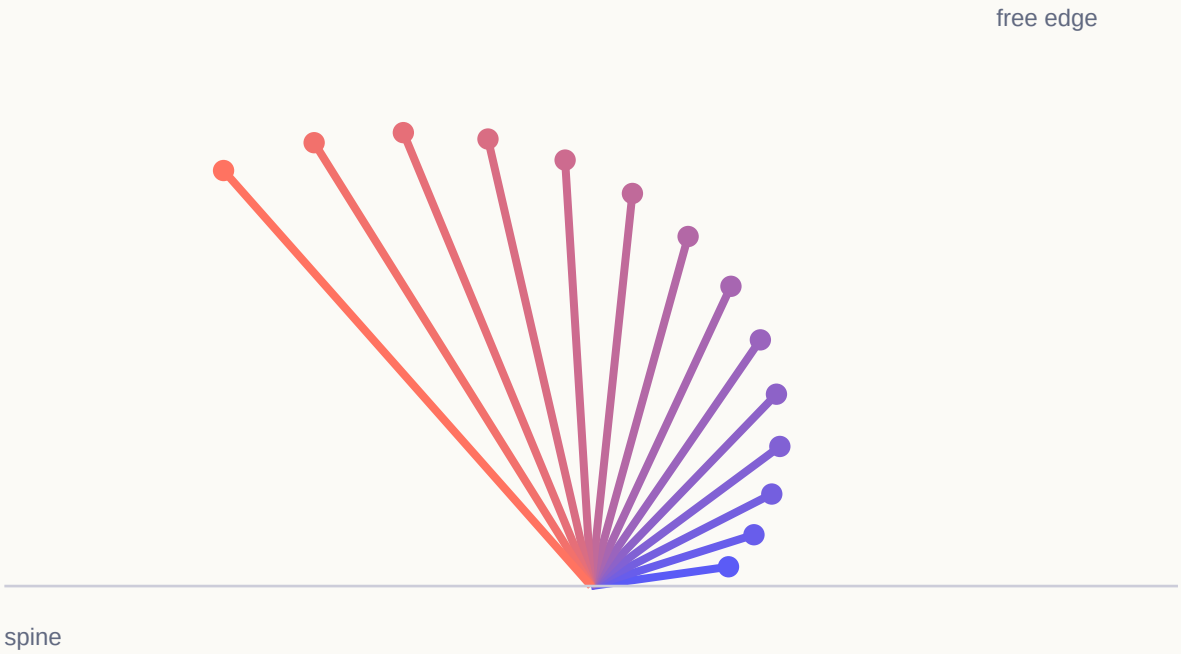
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CHAPTER 1

Real paper, rendered

A turning page in SdFlip is not a flat card that spins on its spine. It is built from dozens of thin vertical strips nested inside each other. Every strip is tilted a few degrees more than its neighbour, so the sheet bends in a smooth arc: the free edge leads, the spine follows, exactly like a real page lifted by its corner.

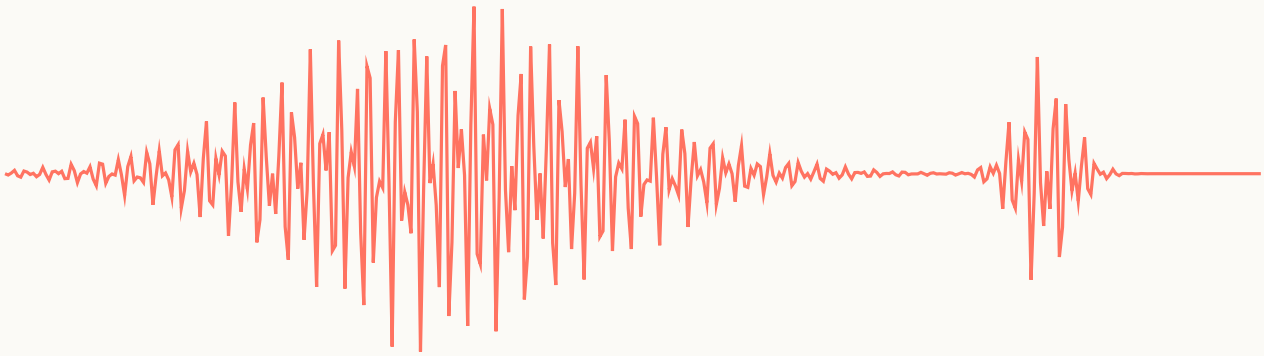


Light comes from the front. The more a strip turns away from you, the darker its gradient gets, and the page casts a soft shadow on the sheet below. The whole effect is CSS 3D transforms, so the browser hands it to the GPU and it runs at full frame rate, even on a phone.

CHAPTER 2

Sound you can feel

There are no audio files to download. SdFlip synthesizes the sound of paper with the Web Audio API: a whoosh of air under the sheet, the rustle of its surface, a crackle as it bends, and a soft slap when it lands.



rustle land

Several variants are rendered ahead of time in the background, so a flip only has to play a buffer. A fast flip sounds shorter and brighter than a slow one. Prefer your own recording? Pass soundUrl and SdFlip plays that file instead.

CHAPTER 3

Made for thumbs

- **Swipe**
  Drag the page with a finger: the curl follows your thumb and finishes the turn when you let go.

- **Pinch**
  Two fingers zoom around the point between them, and pages re-render sharp at the new size.

- **Double tap**
  Zooms in on the spot you tapped. Another double tap zooms back out.

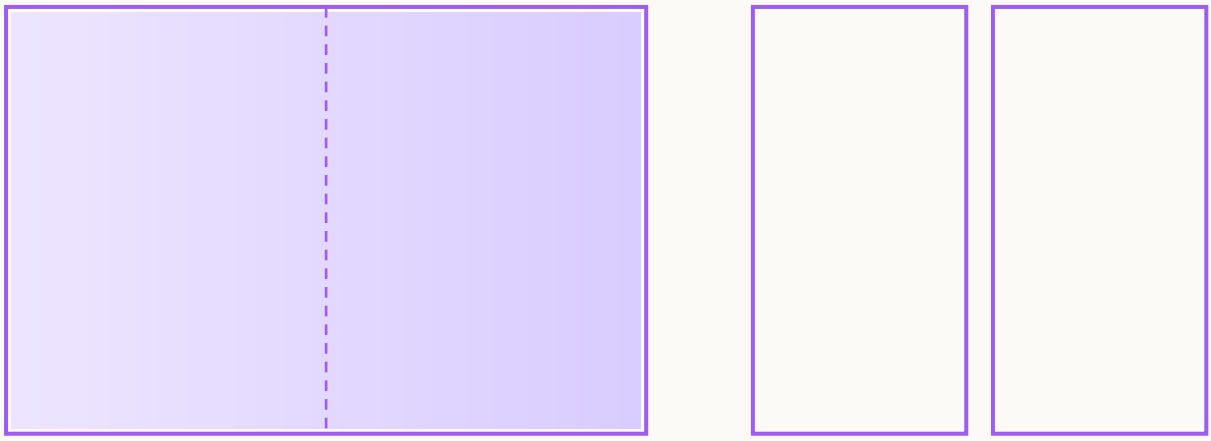
- **Rotate**
  Portrait shows one page, landscape shows a spread. The open page stays open.

- **Scroll**
  An inline viewer never traps the page: vertical swipes scroll the site, horizontal ones flip.

CHAPTER 4

Spreads without the stretch

Many catalogs are exported as double-width spreads. A naive viewer squeezes each one into a single page. SdFlip notices the format, splits every spread into a left and a right half, and puts the book back together so the spread opens exactly as the designer laid it out.



one PDF page, two book pages

left

right

Pages of a different size are fitted whole, centered with margins, never stretched. If a spread would start on the wrong side, a blank page keeps the rest aligned. Landscape slides switch the viewer to single pages.

CHAPTER 5

Search, outline and links

The side panel has three tabs: page thumbnails, the outline of the PDF, and full-text search with highlighted snippets. Try it: search for the tall tower that guides ships at night - it appears on one page only.

Links inside the PDF stay clickable. Each line below is a real link:

Visit the project on GitHub

Write an email (mailto link)

Call a number (tel link)

Jump to "Options at a glance"

CHAPTER 6

Fast on slow phones

0 ms to start a flip: the next sheet is assembled while the screen is still.

3 sheets kept in memory around the open page. Long catalogs never crash a phone tab.

1x canvas per page. The page you see is the rendered canvas itself, not a copy.

60 fps target. Rendering pauses while a page flies and resumes when the browser is idle.

Somewhere a lighthouse keeper reads a manual like this one, flipping pages by lamplight.

CHAPTER 7

Accessible by default

The viewer is a dialog with a proper label. Every button has a name and a keyboard shortcut. A polite live region tells screen readers which pages are open. Closing the lightbox returns focus to the link that opened it. People who ask their system for reduced motion get gentler transitions.

Keyboard: Left and Right arrows flip, Home and End jump, + and - zoom, F toggles fullscreen, S toggles sound, T opens thumbnails and Esc closes.

CHAPTER 8

Drop it into any site

Two files and one attribute. No build step, no framework, no jQuery.

```
<link rel="stylesheet" href="sdflip.min.css">
<script src="sdflip.min.js"></script>

<a href="catalog.pdf" data-sdflip>Open the catalog</a>

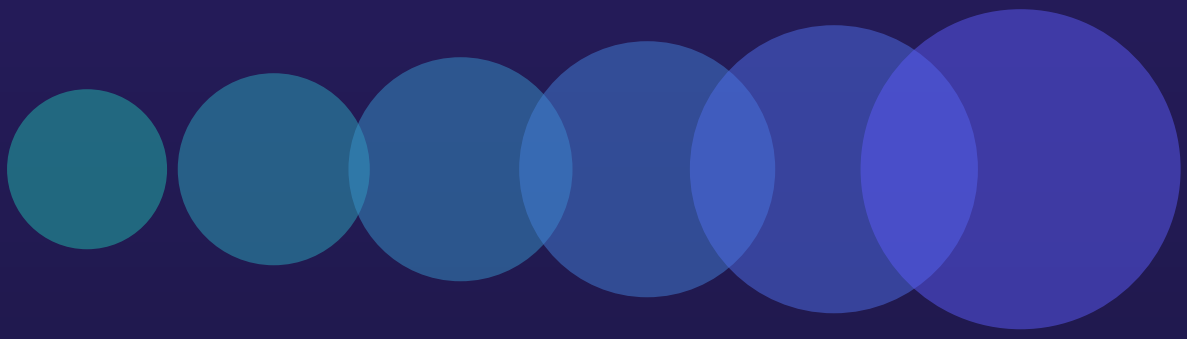
<div id="book" style="height:80vh"></div>
<script>
  SdFlip.inline('#book', { source: 'catalog.pdf' });
</script>
```

pdf.js is loaded on demand from a CDN. To host it yourself, set SdFlip.defaults.pdfjs and SdFlip.defaults.pdfjsWorker, or define window.SdFlipConfig before the script loads.

CHAPTER 9

Options at a glance

`source`	URL, File, Blob or bytes
`title`	caption in the header
`page`	start page
`layout`	auto, double or single
`sound`	page flip sound on or off
`duration`	flip length, ms
`bend`	curl strength, 1 to 2.2
`peel`	corner lifts on hover
`links`	clickable PDF links
`lang`	en, ru or your own locale
`download`	download button
`onFlip`	callback on every turn



Thanks for flipping through.

github.com/Pandarens/SdFlip